Isaac Malsky

isaacmalsky@gmail.com
imalsky
Google Scholar

Appointments

2024–present **Postdoctoral Researcher**, *NASA Jet Propulsion Laboratory*, Pasadena, California
NASA Postdoctoral Program Fellow, Science Division

2019–2024 **Graduate Researcher**, *University of Michigan*, Ann Arbor, Michigan
Rauscher Lab

2015–2019 **Research Assistant**, *University of Chicago*, Chicago, Illinois
Rogers Lab

2018–2019 **Post-baccalaureate Researcher**, *Argonne National Laboratory*, Lemont, Illinois
Chan Lab, Center for Nanoscale Materials

Education

2019–2024 **Ph.D., Astronomy and Astrophysics**, *University of Michigan*, Ann Arbor, Michigan
Advisor: Prof. Emily Rauscher.
Dissertation: *Modeling the 3D Atmospheric Structure of Hot Gaseous Planets*.

2014–2018 **B.A., Physics, with Honors**, *University of Chicago*, Chicago, Illinois
Thesis: *The Coupled Thermal and Compositional Evolution of Planet Envelopes*.

Publications

5 published first-author papers, 2 in review, 838 total citations, and an ADS h-index of 17.

First Author

Malsky, I., Zhang, X., Kataria, T., Graham, M., et al. Accelerating Chemical Kinetics for Exoplanet Atmospheres using Neural Networks. Submitted to *PNAS*.

Malsky, I. and Kataria, T. Accelerating Radiative Transfer for Planetary Atmospheres by Orders of Magnitude with a Transformer-Based Machine Learning Model. Submitted to *ApJ*.

Malsky, I., Rauscher, E., Stevenson, K., et al.. Clouds and Hazes in GJ 1214 b's Metal-rich Atmosphere. (2025). *AJ*.

Malsky, I., Rauscher, E., Roman, M. T., et al.. A Direct Comparison between the Use of Double Gray and Multiwavelength Radiative Transfer in a General Circulation Model with and without Radiatively Active Clouds. (2024). *ApJ*.

Malsky, I., Rogers, L., Kempton, E. M. R., et al.. Helium-enhanced planets along the upper edge of the radius valley. (2023). *Nature Astronomy*.

Malsky, I., Rauscher, E., Kempton, E. M. R., et al.. Modeling the High-resolution Emission Spectra of Clear and Cloudy Nontransiting Hot Jupiters. (2021). *ApJ*.

Malsky, I., Rogers, L. A.. Coupled Thermal and Compositional Evolution of Photoevaporating Planet Envelopes. (2020). *ApJ*.

Contributing Author

Frazier, R. C., Rauscher, E., et al. (incl. **Malsky, I.**). The Days Drag On on WASP-121 b: Interpreting its NIRISS Spectroscopic Phase Curve with General Circulation Models. (2026). *arXiv e-prints*.

Mang, J., Batalha, N. E., et al. (incl. **Malsky, I.**). PICASO 4.0: Clouds and Photochemistry in Climate Models of Brown Dwarfs and Exoplanets. (2026). *ApJ*.

Gardner, T., Monnier, J. D., et al. (incl. **Malsky, I.**). Surprise Nondetection of Upsilon Andromedae b with MIRC-X and MYSTIC at the CHARA Array. (2026). *AJ*.

Gao, P., Piette, A. A. A., et al. (incl. **Malsky, I.**). The Hazy and Metal-rich Atmosphere of GJ 1214 b Constrained by Near- and Mid-infrared Transmission Spectroscopy. (2026). *ApJ*.

van Sluijs, L., Rauscher, E., et al. (incl. **Malsky, I.**). First Results from WINERED: Detection of Emission Lines from Neutral Iron and a Combined Set of Trace Species on the Dayside of WASP-189 b. (2025). *AJ*.

van Sluijs, L., Beltz, H., **Malsky, I.**, et al.. Assessing Robustness and Bias in 1D Retrievals of 3D Global Circulation Models at High Spectral Resolution: A WASP-76 b Simulation Case Study in Emission. (2025). *ApJ*.

Triantafillides, A., Savel, A. B., et al. (incl. **Malsky, I.**). Out on a Limb: The Signatures of East-West Asymmetries in Transmission Spectra from General Circulation Models. (2025). *ApJ*.

Steinrueck, M. E., Parmentier, V., et al. (incl. **Malsky, I.**). The Radiative Effects of Photochemical Hazes on the Atmospheric Circulation and Phase Curves of Sub-Neptunes. (2025). *ApJ*.

Davenport, B., Kennedy, T., et al. (incl. **Malsky, I.**). An Analysis of Spitzer Phase Curves for WASP-121b and WASP-77Ab. (2025). *AJ*.

Kennedy, T. D., Rauscher, E., **Malsky, I.**, et al.. Radiatively Active Clouds and Magnetic Effects Explored in a Grid of Hot Jupiter GCMs. (2025). *ApJ*.

Smith, P. C. B., Sanchez, J. A., et al. (incl. **Malsky, I.**). The Roasting Marshmallows Program with IGRINS on Gemini South. II. WASP-121 b has Superstellar C/O and Refractory-to-volatile Ratios. (2024). *AJ*.

Monnier, J. D., Gardner, T., et al. (incl. **Malsky, I.**). Precision interferometry with MIRC-X/MYSTIC for exoplanets. (2024). *Optical and Infrared Interferometry and Imaging IX*.

Hammond, M., Bell, T. J., et al. (incl. **Malsky, I.**). Two-dimensional Eclipse Mapping of the Hot-Jupiter WASP-43b with JWST MIRI/LRS. (2024). *AJ*.

Bell, T. J., Crouzet, N., et al. (incl. **Malsky, I.**). Nightside clouds and disequilibrium chemistry on the hot Jupiter WASP-43b. (2024). *Nature Astronomy*.

King, G. W., Corrales, L. a. R., et al. (incl. **Malsky, I.**). The XUV-driven escape of the planets around TOI-431 and nu2 Lupi. (2024). *MNRAS*.

Kempton, E. M. R., Zhang, M., et al. (incl. **Malsky, I.**). A reflective, metal-rich atmosphere for GJ 1214b from its JWST phase curve. (2023). *Nature*.

Beltz, H., Rauscher, E., Kempton, E. M. R., **Malsky, I.**, et al.. Magnetic Effects and 3D Structure in Theoretical High-resolution Transmission Spectra of Ultrahot Jupiters: the Case of WASP-76b. (2023). *AJ*.

Murphy, M. M., Beatty, T. G., Roman, M. T., **Malsky, I.**, et al.. A Lack of Variability between Repeated Spitzer Phase Curves of WASP-43b. (2023). *AJ*.

Savel, A. B., Kempton, E. M. R., et al. (incl. **Malsky, I.**). Diagnosing Limb Asymmetries in Hot and Ultrahot Jupiters with High-resolution Transmission Spectroscopy. (2023). *ApJ*.

Christie, D. A., Lee, E. K. H., et al. (incl. **Malsky, I.**). CAMEMBERT: A Mini-Neptunes General Circulation Model Intercomparison, Protocol Version 1.0.A CUISINES Model Intercomparison Project. (2022). *PSJ*.

Beltz, H., Rauscher, E., Kempton, E. M. R., **Malsky, I.**, et al.. Magnetic Drag and 3D Effects in Theoretical High-resolution Emission Spectra of Ultrahot Jupiters: the Case of WASP-76b. (2022). *AJ*.

Gardner, T., Monnier, J. D., et al. (incl. **Malsky, I.**). A thesis to probe unique exoplanet regimes with micro-arcsecond astrometry and precision closure phases at CHARA and VLTI. (2022). *Optical and Infrared Interferometry and Imaging VIII*.

Caldiroli, A., Haardt, F., et al. (incl. **Malsky, I.**). Irradiation-driven escape of primordial planetary atmospheres. II. Evaporation efficiency of sub-Neptunes through hot Jupiters. (2022). *A&A*.

Belkovski, M., Becker, J., Howe, A., **Malsky, I.**, et al.. A Multiplanet System's Sole Super-puff: Exploring Allowable Physical Parameters for the Cold Super-puff HIP 41378 f. (2022). *AJ*.

Caldiroli, A., Haardt, F., et al. (incl. **Malsky, I.**). Irradiation-driven escape of primordial planetary atmospheres. I. The ATES photoionization hydrodynamics code. (2021). *A&A*.

Harada, C. K., Kempton, E. M. R., et al. (incl. **Malsky, I.**). Signatures of Clouds in Hot Jupiter Atmospheres: Modeled High-resolution Emission Spectra from 3D General Circulation Models. (2021). *ApJ*.

Kasper, D., Bean, J. L., Oklopčić, A., **Malsky, I.**, et al.. Nondetection of Helium in the Upper Atmospheres of Three Sub-Neptune Exoplanets. (2020). *AJ*.

Software

Savel, A., M.-R. Kempton, E., Beltz, H., **Malsky, I.**. arjunsavel/hires-literature. (2024). *Zenodo*.

Malsky, I., Cowherd, M., Rauscher, E., et al.. emily-rauscher/RM-GCM: First public version of RM-GCM. (2023). *Zenodo*.

Grants and Awards

2024–2026	NASA Postdoctoral Program Fellowship
2023–2024	Rackham Predoctoral Fellowship
2023	Rackham Professional Development Grant
2023	Rackham Graduate Student Research Grant
2021	NSF Graduate Research Fellowship Program Honorable Mention
2021	Michigan Space Grant
2015	Jeff Metcalf Fellowship
2014	National Merit Scholarship
2014–2018	University of Chicago Dean's List

Mentoring

2025–present	Yashnil Mohanty, high school project
2024–2025	Elijah Mullens, graduate research
2023–2025	Thomas Kennedy, graduate research
2022–2024	Alex Wingate, undergraduate research
2022–2024	Eli Cinque, undergraduate research

Teaching

Spring 2021	ASTRO 105: Cosmos Constellations, Graduate Student Instructor, University of Michigan
Fall 2020	ASTRO 101: Introductory Astronomy: The Solar System and the Search for a New Earth, Graduate Student Instructor, University of Michigan

Service, Outreach, and Skills

Peer review	*The Astrophysical Journal*; *The Astrophysical Journal Letters*; *The Journal of Open Source Software*.
Service	Intro2Astro Course Volunteer Summer (2021) Scarlett Middle School Teaching Assistant (2020) Science Olympiad Mentor at Slauson Middle School (2020) Ann Arbor Astronomy on Tap (2019) Invisible Institute Volunteer Infographic Designer (2018).
Training	Exoclimes Summer School (2023) SC23 Supercomputing Workshops (2023) University of Chicago Supercomputer Workshops (2018).
Computing	Python, FORTRAN 77, FORTRAN 95, git, HPC systems. Basic: C, OpenMP, Julia, SQL.

Observing Proposals

JWST **[Co-I]** Cloudy with a Chance of Disequilibrium Chemistry: A Comprehensive View of Silicate Clouds in a Hot Jupiter with a MIRI/LRS Phase Curve, Splinter et al., GO 11301, Cycle 5, 69.6 hours.

JWST **[Co-I]** Meeting their Match: Measuring Cloudy Hot Jupiter Counterparts to Understand the Silicate Cloud Regime with JWST/MIRI, Moran et al., GO 7849, Cycle 4, 46.3 hours.

Magellan **[Co-I]** The Beginning of a High-Resolution Emission Spectroscopy Program at Magellan for the Characterization of Hot Jupiter Atmospheres with the WINERED Instrument, Sluijs et al., 3 nights (2023).

Gemini South **[Co-I]** Tracing the Day-Night Structure of WASP-121b with Multi-Phase High-Resolution Spectroscopy, Rauscher et al., 12 hours (2023).

Gemini South **[Co-I]** Tracing the Day-Night Structure of WASP-76b with Multi-Phase High-Resolution Spectroscopy, Rauscher et al., 20 hours (2023).

Magellan **[Co-I]** The First High-Resolution Emission Spectrum of an Exoplanet with WINERED at Magellan, Rauscher et al., 12 hours (2022).

Selected Presentations

2026 **Yuk Yung Lunch Talk.**

2026 **Montana State University Colloquium.** AI for Exoplanet Modeling/Analysis. (invited talk)

2026 **MIT Monday Afternoon Talks.** AI for Exoplanet Modeling/Analysis. (invited talk)

2026 **ExoPAG 33.** AI for Exoplanet Modeling/Analysis. (invited talk)

2025 **ExSoCal.** Breaking the Accuracy-Speed Trade-off in Radiative Transfer Calculations with Transformer-Based Emulation. (talk)

2025 **Aspen Rocky Worlds Conference.** Clouds and Hazes in GJ 1214 b's Metal-Rich Atmosphere. (talk)

2025 **NASA AMES.** Machine Learning Emulation of Atmospheric Radiative Transfer. (talk)

2024 **Cornell.** Clouds and Hazes in GJ 1214b's Metal-Rich Atmosphere. (talk)

2024 **NASA Jet Propulsion Laboratory.** Characterizing Exoplanet Atmospheres: 1D and 3D. (talk)

2024 **Los Alamos National Laboratory.** Characterizing Exoplanet Atmospheres: 1D and 3D. (talk)

2023 **Exoclimes VI.** Clouds and Hazes in Polychromatic General Circulation Models. (talk)

2022 **Bay Area Exoplanets Meeting.** Clouds and Hazes in Polychromatic GCMs. (talk)

2022 **The Other Worlds Laboratory.** A Direct Comparison between Double Gray and Multiwavelength GCMs. (talk)

2021 **Great Lakes Exoplanet Area Meeting.** Modeling the Three-Dimensional Atmospheric Structure of Hot Gaseous Planets. (talk)

2020 **Exoplanets IV.** Modeling the High-resolution Emission Spectra of Clear and Cloudy Nontransiting Hot Jupiters. (poster)

2018 **MIT-Harvard Poster Session.** The Coupled Thermal Evolution of Sub-Neptune Atmospheres. (poster)

2018 **The US Naval Academy.** The Coupled Thermal Evolution of Sub-Neptune Atmospheres. (poster)